Exact Correlated Readouts and a Minimal Class Lift in a Finite WRRA Model

A computational model-development and verification note

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20 October 2026 | v1.1
Release: WRRA-CONTRACT-AUDIT-20261020-v1.1
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Abstract

We present a reproducible interface audit for a finite classical-quantum WRRA model. Three sector-weighted carrier operators preserve the declared snapshot carrier queries but do not close under an address-controlled update. Using actual fixed coefficients on 256 active addresses, two positive state families have maximum initial sector Z-readout difference 1.11 x 10^-16. An identity/bit-flip update separates their return-sector Z readouts by 0.6668017457457079, while class-resolved weighted operators repair the update to 1.11 x 10^-16. Exact triangular witnesses establish coefficient ranks three and six. We then distinguish two update contracts: arbitrary within-class channels require the full class-query representation, whereas the prescribed qubit I/X involution admits a smaller exact representation with 15 variable real coordinates instead of 18. A reusable coefficient/mask compiler constructs the reduced state, transition and readout maps; an eight-step trajectory agrees to 1.82 x 10^-15. The contribution is a model-specific contract comparison and executable reduction artifact using established mathematics. No new gravitational law, particle spectrum or universal speedup is claimed.

Assessment contract

Verification input: fixed finite address weights, admission coefficients, normalized conditional states and specified control classes.
WRRA transformation: address-conditioned carrier states become weighted readout operators; the actual class mask tests dynamical closure and determines a sufficient lift.
Output: a numerical failure witness, exact rank witness, restored readouts and explicit storage contracts.
Falsification: failure of positivity, equal-initial-state conditions, direct-versus-reduced identities, the stated class-lift update or the exact rank witnesses rejects the implementation claim. Physical interpretation requires a separately specified empirical mapping.

1. Purpose and relation to established reduction methods

WRRA means Worldline-Residue-Resource-Action. This note concerns its finite address/carrier interface. The practical question is which state representation an implementer may safely retain when moving from fixed readouts to address-controlled evolution. Preserving a snapshot is insufficient grounds for discarding the address classes needed by future updates.

Observable-algebra reduction [1] and invariant-space methods for exact quantum dynamical reduction [2] already supply the general mathematical framework. We do not present the weighted-sum identity, the linear rank bound or invariant-subspace closure as new general theorems. Their role here is executable diagnosis: a fixed WRRA interface fails, a concrete null-space witness exposes the lost distinction, an explicit class lift repairs it, and exact minors certify the required dimension for a specified query family. This is a model-specific computational case study, not a competing general reduction algorithm.

The parent paper already documented the failure witness and full class lift. The new reusable artifact here accepts arbitrary finite coefficient matrices and binary control masks, compiles the exact fixed-qubit I/X observable orbit, and returns encoding, transition and readout matrices. The 15-versus-18 contract comparison is added here rather than attributed to the parent result. It is useful to implementers of finite classical-quantum mixtures with this fixed control, whether their coefficients are arithmetic or supplied by another model. We claim this implementation and WRRA application, not priority for the general observable-orbit construction.

2. State and snapshot contract

```
Omega = sum_n w_n |n><n| tensor rho_n
w_n > 0; sum_n w_n=1; rho_n >= 0; Tr(rho_n)=1
Sigma_s = sum_n w_n f_s(n) rho_n; s=P,D,R
query(s,B) = Tr(Sigma_s B)
```

The address register is classical. Each conditional carrier state is a d-dimensional density operator. The supplied nonnegative f_s are response coefficients, not necessarily probabilities. Every Hermitian carrier matrix B is an allowed query for each sector. Conditional states need not be mutually commuting. Positivity of each weighted operator follows termwise; linearity gives exactly the same query as summing the full address-conditioned state. Fixed scalar readouts require only their scalar answers; the larger arbitrary-B contract requires the weighted operators.

Let F have entries f_s(n), rank r, and fix strictly positive weights. Expand each trace-zero Hermitian perturbation of rho_n in d^2-1 basis matrices. In each direction the output perturbation is F diag(w) x, which has image dimension r. Small perturbations around I/d remain positive, so all r(d^2-1) directions are attainable. An exact linear encoding for this full query family therefore requires that many variable real coordinates, and r independent weighted operators attain the bound. Known trace parts are fixed and are not variable coordinates. This statement does not bound arbitrary nonlinear physical implementations.

3. Why separate marginals can fail

```
L_s - mu_s Tr(rho_bar A_s) = Cov_w(f_s,b_s)
mu_s=sum_n w_n f_s(n); rho_bar=sum_n w_n rho_n
b_s(n)=Tr(rho_n A_s)
```

Replacing the joint address/carrier state by its separate marginals removes precisely this covariance. The identity is elementary; its implementation consequence is that matching both marginals does not guarantee matching the modeled readouts. The supplied script checks the identity on actual coefficients. This is an internal explanation of lost information, not evidence that a particular correlation has been observed in nature.

4. Address-controlled evolution and exact repair

Partition the address set into fixed disjoint classes C_k. Keep weights and response coefficients fixed. Carrier states in class k evolve by a common completely positive trace-preserving map Phi_k. Define class-resolved operators Sigma_sk=sum over n in C_k of w_n f_s(n) rho_n. By linearity, the exact updated readout operators are

```
Sigma_s' = sum_k Phi_k(Sigma_sk).
```

For the complete class-resolved arbitrary-carrier query family, replace F by the stacked class-masked rows F_lift=(f_s 1_Ck). The same rank argument gives r_lift(d^2-1) variable real coordinates. This sufficient representation supports arbitrary sequences of the declared within-class common maps. Changing weights, response coefficients, classes or mixing addresses changes the contract and requires a new audit.

For arbitrary independently selectable class maps, closure of the original operator family is equivalent to invariance of the row space of F under each class mask. Sufficiency follows because each masked weighted operator is then a known linear combination of retained operators. For necessity when d is at least two, a masked row outside that row space has a component in the original coefficient null space. Opposite sufficiently small traceless conditional-state perturbations are invisible initially. Choosing different identity/bit-flip channels on the relevant class distinguishes them. Thus no universally closed original-interface update can exist for that class-map family. This is the established invariant-space principle specialized to the present coefficients.

5. Exact WRRA rank witness

The address domain is n=2,...,1,015,000. A supplied phase-admission effect e_P is positive on odd composites and zero elsewhere. e_D is one on even composites and zero elsewhere; e_R=1-e_P-e_D. Response coefficients are f_s=e_s[1+lambda_s log(n)/log(N)], with lambda=(0,1/4,1/10). The two classes are n divisible by three and its complement.

The fixed address measure is w_n=n^(-alpha)/sum_m m^(-alpha), alpha=1.8996877935161325. On odd composites, e_P=1-product over k=0,...,7 of [1-sigmoid(d_k-h)], h=1.4476744689338703, sigmoid(x)=1/(1+exp(-x)), and d_k=sum over j=1,...,4 of cos[gamma_j(log n+0.1k)]/2. The supplied gamma values are 14.134725141734695, 21.022039638771556, 25.01085758014569 and 30.424876125859512. The measure, spectral values and threshold are construction inputs. No parameter is fitted in the present rerun.

Control class	Three witness addresses
Divisible by three	3 (prime), 6 (even composite), 9 (odd composite)
Complement	2 (prime), 4 (even composite), 25 (odd composite)

Within either class, the prime column has only a positive R entry, the even-composite column has only a positive D entry, and the odd-composite column has a strictly positive P entry. The determinant is nonzero by a triangular permutation, independently of numerical thresholds. Thus each class contributes rank three. Disjoint class support makes the stacked rank exactly six; the original three-row matrix has rank three. For d=128 this yields 3(128^2-1)=49,149 snapshot coordinates and 6(128^2-1)=98,298 complete class-resolved coordinates. Six blocks are not asserted minimal for replaying only one prescribed I/X update or only three fixed scalar outputs.

6. Reproduced failure using the actual coefficients

Use the first 256 active addresses n=2,...,257. Their weights are inherited from the full normalized address measure rather than renormalized for the witness. Set A_sn=w_n f_s(n) on this subset and z_n=+1 when n is divisible by three, otherwise -1. Define

```
u_n=A_Rn z_n
h=u-A^T (A A^T)^(-1) A u; delta=h/max_n |h_n|
rho_n^(+/-)=(I +/- delta_n Z)/2
```

Assign I/2 to all remaining addresses in both alternatives. Since A delta=0, every initial weighted operator agrees, not merely the displayed Z query. Since |delta_n| is at most one, both conditional-state families are positive and trace one. Apply I on the divisible class and X on its complement. XZX=-Z, so the post-update difference is the explicit vector 2 A diag(z) delta. The class-resolved sums preserve the missing distinction and reproduce that vector. The qubit construction embeds in the 128-dimensional carrier by restricting to a two-dimensional subspace.

Recomputed diagnostic	Result
Maximum across the three initial sector Z residuals	1.110e-16
Post-update P-sector Z difference	4.5366739244e-05
Post-update D-sector Z difference	0
Post-update R-sector Z difference	0.6668017457457079
Maximum class-lift repair error	1.110e-16
20 sampled Hermitian-query preservation tests: maximum error	4.441e-16
Covariance identity error	2.212e-17

Eighteen declared checks pass, covering the original witness and the added fixed-update compiler. These checks overlap and are not eighteen independent observations. The reported initial residual is specifically max_s |2 A_s delta| across the three Z readouts, not a supremum over unrestricted Hermitian matrices. The rank proof is analytic; floating-point determinant checks only confirm its two explicit witnesses numerically. The 0.6668 difference reproduces the documented parent construction and is not a newly measured quantity.

7. Practical implementation decision

Form the query coefficient matrix; determine the retained row space; mask it by each planned control class; test rank growth; if it grows, construct a null-space witness and retain sufficient class-resolved blocks before executing updates. The present case requires a class lift for the full class-query contract. A common carrier channel with no address control needs no lift. These tests prevent a valid snapshot cache from silently becoming an invalid simulator.

8. A smaller exact representation for the prescribed I/X update

The full class lift handles arbitrary independently selected channels. A fixed qubit I/X update is a narrower contract. Write every conditional state as (I+x_n X+y_n Y+z_n Z)/2. Identity contributions are known fixed traces. Let J be the sign mask +1 on the I class and -1 on the X class. One step maps X components to themselves and Y,Z components to J times themselves. Because the update is an involution, all-time readout coefficients lie in the span of the initial coefficients and their one-step images.

```
r = rank(F); r_lift = rank([F; F diag(J)])
dim_fixed_IX = r + 2 r_lift
dim_full_class_qubit = 3 r_lift
```

Proof and minimality. X-component readouts need the r-dimensional row space of F. Y and Z each need the joint row space of F and F diag(J), whose dimension is r_lift because class-masked rows are (F +/- F diag(J))/2. The three carrier directions are independent, and small Bloch perturbations around the maximally mixed state preserve positivity. Their dimensions therefore add and give a lower bound for linear encodings preserving every sector carrier query at all integer steps of this fixed update. The invariant row-space encoding below attains it. This bound concerns trace-one conditional states, with fixed identity readouts; it does not concern nonlinear encodings or arbitrary class maps.

For the actual WRRA coefficients, r=3 and r_lift=6. The fixed update therefore needs 15 variable coordinates, while the full qubit class-query family needs 18. This comparison is for a qubit carrier, not a claim that a full d=128 fixed-update model has dimension 15.

reusable_closure.py forms C0=F tensor I_3, represents the Bloch update by a diagonal sign matrix D, and computes an orthonormal row basis H of [C0; C0 D]. It returns reduced state v=H b, transition T=H D H^T, and output map O=C0 H^T. Closure gives H D=T H and readout preservation gives C0=O H. Thus v evolves by T and every original X/Y/Z readout is O v. Arbitrary Hermitian qubit queries follow by linear combination with known identity traces.

Compiler result / worked input	Output
Actual 256-address matrix, fixed I/X	15 coordinates; full class family 18
Eight-step readout error	1.818e-15
Simple F=[[1,1,0,0],[0,0,1,1]], I-mask=[1,0,1,0]	r=2, r_lift=4: fixed dimension 10; full class dimension 12

Run python3 reusable_closure.py for the simple example. The compiler accepts finite real F and a matching Boolean mask. Its singular-value cutoff is tolerance times max(1,largest singular value), with default tolerance 10^-12. Near-degenerate numerical ranks require conditioning analysis; the actual rank claim here is separately certified by the structural witnesses. The reusable routine is an implementation of established observable-space reduction [1,2].

9. Resource comparison follows the contract

Representation at d=128, G=32	Retained bytes and capability
Three fixed float64 loads	24; only their fixed scalar answers
32 complex128 pure vectors and 3 x 32 float64 weights	66,304; grouped state and weighted queries
Three complex128 weighted operators	786,432; arbitrary sector carrier queries at the snapshot

These are explicit storage counts excluding working buffers; they are not measured runtime speedups. Dense weighted operators use more storage than this grouped input. The compact grouped form remains advantageous when its classes and pure-state description are available. A complete class-resolved operator family supports the stated controlled maps but costs more. The purpose of the audit is to choose a sufficient representation, not to establish a universal minimum or a universally fastest implementation. Timing data from earlier WRRA work are not represented here as fresh measurements.

10. Supplementary color-sector audit

A separate 36-check audit replays the original component branch and supplied three-quark color singlet. It reproduces 427,892 odd-composite addresses, phenotype share 0.050000000000000065 and component-order TV distance 0.20472100116369085. Original charge assignments and candidate neutron labels remain disclosed inputs.

Source inspection finds that color_check() constructs three colors, the six signed permutations and eight generators internally, consuming no arithmetic state. Its success therefore establishes compatibility with its chosen representation. It cannot demonstrate arithmetic selection of SU(3). Exact symbolic nullspace calculation gives a one-dimensional common kernel on three fundamental color factors, spanned by the usual normalized antisymmetric state |S>. A joint density operator supported on that color sector necessarily has the form sigma_address tensor |S><S|: expanding in address matrix units, every color block is a scalar multiple of the rank-one projector. Hence this restricted color sector carries no independent address-color correlations.

This elementary representation-theory consequence [3] changes another implementation decision: arithmetic dependence needs an explicit larger-state or coupling rule; it cannot be inferred from a successful stand-alone singlet test. Exact singlet support is stronger than zero mean charge. The restriction does not apply unchanged when omitted gluon, flavor or spatial factors are retained. The supplied color supplement contains the proof, negative controls, source-dependency table and full check results. Separately implemented checks within this project are not claims of independent experimental replication.

11. Scope

The contribution is a reproducible finite-model interface failure and repair, with exact coefficient-rank witnesses and a complementary source-dependency restriction. Standard mathematics is credited; no broad theoretical priority is claimed. Verified inputs and calibrated model construction are legitimate here. A new physical prediction is not a prerequisite for this computational note, and none is asserted. Whether WRRA explains measured gravity or particle properties remains a separate question requiring the corresponding internally implemented dynamics and physical comparison.

12. Reproduction and release identity

The companion release WRRA-CONTRACT-AUDIT-20261020-v1.1 contains this manuscript, contract_audit.py, reusable_closure.py, audit.py, eight frozen upstream source files, SOURCE_SHA256.json, contract_results.json, audit_results.json, the color supplement and version-specific AI review records. RELEASE.json content-identifies the final manuscript and executable files with SHA-256 values. The upstream repository commit is 9d9daf21154e46e55b9fd0fd461c7c070156ab94. All eight source-file bytes match its Git blob hashes. No new public DOI is claimed; the complete companion package is supplied with this note.

```
python3 -m venv .venv
.venv/bin/python -m pip install numpy==2.3.5 scipy==1.17.0 sympy==1.14.0
.venv/bin/python contract_audit.py
.venv/bin/python audit.py
```

Recorded runtime: Python 3.12.14. The scientific scripts need NumPy, SciPy and SymPy; ReportLab is needed only to rebuild PDFs. Source manifests are checked before scientific execution. Changing a frozen source makes the audit stop. Rerunning writes the corresponding results JSON; differing numerical libraries or version metadata can change its release hash. Preserve distributed reference outputs for byte comparisons and use declared numerical tolerances for scientific comparisons. Matrix/readout tolerances are 10^-12, historical return-difference tolerance is 2 x 10^-12, numerical rank threshold for the color audit is 10^-10, and exact symbolic identities have zero tolerance. The address-rank argument uses exact triangular structure rather than a floating-point rank claim.

The targeted rerun extracts original address/admission function bodies by Python AST and supplies their stated globals. It bypasses unrelated import-time cosmology modules and is not the full original Stage-3 pipeline. contract_audit.py independently codes the explicit witness already documented in the parent paper, and compares its main output against that reference. Random diagnostics use seed 20261020. All software-success statements in this note refer to the supplied tests; no laboratory or observational validation is implied.

References

[1] O. Kabernik, J. Pollack and A. Singh, Quantum State Reduction: Generalized Bipartitions from Algebras of Observables, Physical Review A 101, 032303 (2020). DOI: 10.1103/PhysRevA.101.032303.

[2] T. Grigoletto, Y. Tao, F. Ticozzi and L. Viola, Exact Model Reduction for Continuous-Time Open Quantum Dynamics, Quantum 9, 1814 (2025). DOI: 10.22331/q-2025-07-29-1814. This develops general reduction machinery; the present discrete class-mask audit is a bounded application, not a replacement for that theory.

[3] D. Tong, The Standard Model, chapter 3, The Strong Force, University of Cambridge lecture notes, equations (3.4)-(3.6) and (3.95). https://davidtong.org/pdfs/teaching/standard-model/standardmodel3.pdf. Accessed 20 October 2026.

[4] W. Choi and J. Choi, WRRA computational sources and parent manuscript, repository https://github.com/Wonsik-Choi-janefather/wrra-m-0.1 at commit 9d9daf21154e46e55b9fd0fd461c7c070156ab94. Parent paper v1.8 documents the actual-interface witness. The original Stage-3 branch and calculations/wrra_m_0_9 supply the frozen admission and component functions.